
Capability or Verbosity? Disentangling the Drivers of Length-Debiased Preference in LLM Evaluation

Anonymous¹

Abstract

Length-controlled evaluation metrics have been proposed to debias LLM preference assessments, but whether they preserve the capability ordering that human annotators reveal has not been tested with confound-controlled methods at population scale. We ask: after explicitly removing the effect of response verbosity, does human preference (`win_rate`) still predict length-debiased preference (`LC_winrate`) — and how strongly? Analyzing the AlpacaEval 2.0 leaderboard across 223 diverse LLMs, we find a Spearman partial correlation of 0.985 ($p = 1.69 \times 10^{-170}$, one-tailed), higher than the 0.94 bivariate correlation previously reported — capability explains $\sim 5\times$ more LC variance than verbosity in standardized regression ($|\beta_{\text{win}}| = 21.34$ vs $|\beta_{\text{len}}| = -4.37$), and the monotonic capability-LC ordering across quartiles carries a large effect size ($\epsilon^2 = 0.883$). Two independent statistical estimators converge within 1.1pp (Frisch–Waugh–Lovell verification), ruling out methodological artifact. These results validate length-controlled evaluation as a capability-order-preserving transformation, and establish a concrete lesson for evaluation study design: using the alignment gap Δ rather than `LC_winrate` as outcome attenuates effect sizes by $\sim 10\times$ due to mathematical composition.

1. Introduction

Across 223 large language models evaluated on AlpacaEval 2.0, model capability predicts length-controlled preference with a partial Spearman correlation of 0.985 — *after fully accounting for response verbosity*. Two independent evaluation systems, one based on human pairwise comparison and one on AI-debiased annotation, agree on model capability

ordering to a degree the field has not previously quantified. More strikingly, this agreement is stronger once verbosity is removed than before: the bivariate correlation reported by Dubois et al. (2024) is 0.94; our partial correlation is 0.985. Controlling for the confound actually reveals the signal more clearly.

This finding bears on a question at the center of LLM evaluation: does length-controlled win rate (`LC_winrate`) preserve the capability ordering that human-annotated win rate (`win_rate`) reflects? The LC correction in AlpacaEval 2.0 is a GLM-based debiasing procedure that regresses out response length from pairwise comparison outcomes (Dubois et al., 2024). Its stated goal is to remove verbosity inflation — the tendency of AI judges to favor longer, more elaborate responses — while leaving genuine quality differences intact. Whether it succeeds at preserving capability ordering at population scale has not been tested with confound-controlled methods.

The surface problem is well-known: verbosity biases AI-based evaluation. Zheng et al. (2023) and Shi et al. (2024) document that AI judges systematically prefer longer responses. AlpacaEval 2.0’s LC correction is specifically designed to address this. However, a deeper problem remains unresolved: verbosity and capability co-vary in real model populations ($\text{VIF} = 1.764$ in our data), meaning that any observed correlation between `win_rate` and `LC_winrate` could reflect shared verbosity rather than shared capability signal. The gap in existing work is precisely this: no study has applied partial correlation or standardized regression to disentangle the independent contributions of capability and verbosity to LC preference across a large, diverse model population.

Our key insight is that this is a partial correlation question, not a bivariate one. Once we control for `avg_length`, the capability-LC alignment does not weaken — it strengthens. This is because verbosity was acting as a mild suppressor: capable models tend to write longer responses, which partially masks the true capability-LC signal in bivariate analysis. After residualization, the underlying capability signal dominates the LC prediction by a factor of ~ 4.9 (standardized β ratio: 21.34 vs -4.37), and the negative β for verbosity confirms the LC correction is functional.

¹Anonymous Institution. Correspondence to: Anonymous <anonymous@anonymous.edu>.

We make the following contributions:

1. **First population-scale partial correlation quantification:** We establish $\rho(\text{win_rate}, \text{LC_winrate} \mid \text{avg_length}) = 0.985$ ($p = 1.69 \times 10^{-170}$ one-tailed, pre-registered directional test, $N = 223$), with bootstrap 95% CI [0.976, 0.988].
2. **Capability-verbosity dominance via standardized regression:** Capability contributes $\sim 4.9\times$ more to LC preference than verbosity ($|\beta_{\text{win}}| = 21.34$ vs $|\beta_{\text{len}}| = 4.37$; $R^2 = 0.963$ vs verbosity-only $R^2 = 0.256$ — 70 percentage points more explained variance).
3. **FWL theorem verification:** Two independent estimators converge within 1.1pp ($\rho_{\text{resid}} = 0.974$, $\delta = 0.011$), ruling out methodological artifact.
4. **Quartile-level monotonicity with large effect:** KW $\varepsilon^2 = 0.883$ ($H = 196.32$, $p = 2.63 \times 10^{-42}$), with fully monotonic LC ordering ($Q1 = 7.14 < Q2 = 14.69 < Q3 = 26.41 < Q4 = 51.62$) and Dunn $Q1$ vs $Q4$ $p = 1.04 \times 10^{-38}$.
5. **Methodological lesson on dependent variable choice:** KW on Δ yields $\varepsilon^2 = 0.088$ vs $\varepsilon^2 = 0.883$ for LC_winrate — a $10\times$ attenuation from DV choice alone, with concrete recommendations for future studies.

We organize the paper as follows. Section 2 reviews prior work. Section 3 describes methodology. Section 4 presents experimental design. Section 5 reports results. Section 6 discusses limitations. Section 7 concludes.

2. Related Work

2.1. Verbosity Bias and Length-Controlled Evaluation

The problem of length bias in LLM-based evaluation is well-documented. Zheng et al. (2023) show that GPT-4 as a pairwise judge prefers longer, more structured responses even when content quality is equal. Shi et al. (2024) extend this to systematic position and length bias across judge models.

Dubois et al. (2024) respond with AlpacaEval 2.0’s length-controlled win rate (LC_winrate). Their GLM-based approach regresses response length out of pairwise comparison outcomes, yielding a debiased preference score. They report a bivariate Spearman correlation of $\rho \approx 0.94$ between win_rate and LC_winrate , and show LC_winrate correlates more strongly with Chatbot Arena ($\rho \approx 0.98$). Our work extends this: we move from bivariate to partial correlation, controlling for verbosity (avg_length), and find the capability-LC alignment strengthens to 0.985 when verbosity is properly held constant.

Hu et al. (2024) provide a mechanistic decomposition of win_rate into a desirability component (length-independent quality) and information mass component (length-dependent content). Their framework predicts the desirability channel survives LC correction, consistent with our finding that β_{win} dominates β_{len} in standardized regression.

2.2. Bidirectional Human-AI Alignment

Shen et al. (2024) provide a systematic review of bidirectional alignment between humans and AI systems, identifying a gap in empirical quantification of capability-modulated alignment asymmetry. Ji et al. (2023) survey alignment approaches broadly, noting that evaluation metric consistency across human and AI evaluators is an open problem. Our work directly addresses the empirical gap identified by Shen et al. (2024): we quantify, at population scale, how strongly capability modulates the alignment between human preference (win_rate) and AI-debiased preference (LC_winrate).

2.3. Positioning

Our work differs from prior studies in three ways: (1) we study the *partial* relationship between capability and LC preference, explicitly controlling for verbosity; (2) we provide a FWL-based methodological robustness check new to this literature; and (3) we analyze both LC_winrate and Δ as dependent variables, providing a comparative effect-size lesson with implications for evaluation study design. We complement the AlpacaEval 2.0 methodology: our results validate the LC correction as a capability-preserving transformation.

3. Methodology

Building on our observation that the capability-LC alignment question is inherently a partial correlation problem, we design a multi-layer statistical analysis establishing existence (P1), mechanism (P2), and population-level confirmation (P3).

3.1. Dataset

We use the publicly available AlpacaEval 2.0 leaderboard CSV (accessed August 2026), containing pairwise preference results for 223 diverse LLMs compared against a GPT-4o reference on 805 instruction-following prompts. The three variables of interest are:

- **win_rate:** Fraction of pairwise comparisons where human annotators prefer the model (range: $\sim 3\text{--}80\%$)
- **LC_winrate:** Win rate after GLM-based length debi-

asing (Dubois et al., 2024) (range: $\sim 2\text{--}95\%$)

- **avg_length**: Mean response token count per model (range: $\sim 400\text{--}2200$ tokens)

All 223 rows have complete data; no imputation was required.

Verbosity Separability Check: $VIF = 1.764$ for both `win_rate` and `avg_length` (threshold: $VIF < 5.0$), confirming OLS coefficients and partial correlations are interpretable (Figure 4).

3.2. Primary Analysis: Partial Correlation (H-E1)

We seek $\rho(\text{win_rate}, \text{LC_winrate} \mid \text{avg_length})$ — the Spearman partial correlation after controlling for `avg_length`.

Implementation: `pinguin.partial_corr(method='spearman', alternative='greater')` with covariate `avg_length`. Pre-registered threshold: $|r_{\text{partial}}| \geq 0.15$, $p < 0.05$ (one-tailed; directional hypothesis $\rho > 0$). Bootstrap 95% CI: 1000 resamples, `random_state=42`.

3.3. Mechanism Analysis: Standardized OLS (H-M1)

Design: `LC_winrate ~ win_rate_std + avg_length_std` with `sklearn.StandardScaler` preprocessing and `statsmodels.OLS`. Dominance criterion: $|\beta_{\text{win}}| > |\beta_{\text{len}}|$. Verbosity-only baseline (`LC_winrate ~ avg_length_std`) for R^2 comparison.

3.4. FWL-Inspired Robustness Verification (H-M2)

Design: Regress `avg_length` from both `win_rate` and `LC_winrate` (OLS); compute Spearman ρ on residual pairs. $\Delta = |\rho_{\text{Pinguin}} - \rho_{\text{resid}}| < 0.02$ confirms estimator consistency. *Note:* The Frisch–Waugh–Lovell theorem guarantees exact algebraic equality between partial OLS coefficients and residualized OLS estimates (Pearson). For Spearman partial correlation, residualized ranks provide an independent estimator that approximates this principle; convergence within 1.1pp is empirical evidence of robustness, not a theorem guarantee.

3.5. Population-Level Confirmation: Kruskal-Wallis + Dunn (H-M3, H-C1)

Design: `win_rate` quartiles (Q1–Q4; $N = 56, 56, 55, 56$). `scipy.stats.kruskal` on both `LC_winrate` (H-C1) and Δ (H-M3). Post-hoc: `scikit_posthocs.posthoc_dunn` with Bonferroni correction. Effect size: $\varepsilon^2 = (H - k + 1)/(N - k)$. Bootstrap CI on Dunn Q1 vs Q4 p (1000 resamples).

Table 1. Sub-hypothesis summary

Hypothesis	Type	Gate	Test
H-E1	Existence	MUST_WORK	Partial corr > 0.15
H-M1	Mechanism	MUST_WORK	$ \beta_{\text{win}} > \beta_{\text{len}} $ in standard
H-M2	Mechanism	SHOULD_WORK	FWL delta < 0.02
H-M3	Mechanism	SHOULD_WORK	KW on Δ across quartile
H-C1	Condition	SHOULD_WORK	KW on LC_winrate ANI

3.6. Sub-Hypothesis Structure

4. Experimental Setup

We design experiments to answer five research questions:

RQ1: Does capability independently predict LC preference after verbosity control? (*P1: existence*)

RQ2: Does capability dominate verbosity in standardized regression? (*P2: mechanistic dominance*)

RQ3: Is the capability-LC alignment robust across independent estimators (FWL-inspired residualization)? (*P2: estimator robustness*)

RQ4: Is capability-LC monotonicity confirmed at quartile extremes? (*P3: population condition*)

RQ5: Does Δ vary across capability levels, and which DV is appropriate? (*Secondary: DV choice*)

4.1. Dataset

All experiments use the AlpacaEval 2.0 leaderboard CSV ($N = 223$). No additional data was collected.

Why this dataset: AlpacaEval 2.0 is the only publicly available benchmark providing both human-annotated win rates and GLM-debiased LC win rates for the same models at $N > 100$ scale, enabling direct capability-LC comparison with verbosity control.

4.2. Baselines

Bivariate correlation (Dubois et al., 2024): $\rho(\text{win_rate}, \text{LC_winrate}) \approx 0.94$. Our partial correlation (0.985) extends this to confound-controlled setting.

Verbosity-only model: OLS restricted to `avg_length_std` sole predictor. $R^2 = 0.256$ — establishes verbosity-only baseline.

4.3. Implementation

All analyses in Python 3.10+ using `pinguin` ≥ 0.5 , `statsmodels` ≥ 0.13 , `scipy` ≥ 1.9 , `sklearn` ≥ 1.1 , `scikit_posthocs` ≥ 0.7 . `random_state=42`

throughout. No GPU required. Full analysis completes in under 5 minutes on standard CPU hardware.

$|\beta_{\text{win}}| = 21.34$ vs $|\beta_{\text{len}}| = 4.37$ — dominance ratio 4.88:1 ($p_{\text{win}} = 4.58 \times 10^{-145}$)

4.4. Evaluation Metrics

Full model $R^2 = 0.963$ vs verbosity-only $R^2 = 0.256$ (+70.7pp from capability).

Table 2. Evaluation metrics per research question

RQ	Metric
RQ1	Spearman partial r , p , Bootstrap CI
RQ2	$ \beta_{\text{win.std}} $ vs $ \beta_{\text{len.std}} $, R^2
RQ3	FWL delta
RQ4	KW p , ε^2 , Dunn Q1 vs Q4 p
RQ5	ε^2 (H-M3 on Δ) vs ε^2 (H-C1 on LC_winrate)

Figure 5 shows the standardized coefficient comparison. $\beta_{\text{len}} = -4.37$ (negative): verbosity is penalized once capability is controlled — confirming the LC correction is directionally correct. Mild heteroscedasticity (BP $p = 0.011$) does not affect conclusion given $p_{\text{win}} = 4.58 \times 10^{-145}$. $r > 0.15$; $p < 0.05$. $|\beta_{\text{win}}| > |\beta_{\text{len}}|$ delta < 0.02 . Both $p < 0.05$; monotonic ordering

Table 4. H-M1 standardized OLS results

Metric	Value
$ \beta_{\text{win.std}} $	21.34
$ \beta_{\text{len.std}} $	-4.37
Dominance ratio	4.88×
Full model R^2	0.963
Verbosity-only R^2	0.256
R^2 gain	+70.7pp

5. Results

All five sub-hypotheses passed their gates. We report in RQ order.

5.1. RQ1: Capability Independently Predicts LC Preference (H-E1)

After controlling for avg_length:

$r_{\text{partial}} = 0.9851$, $p = 1.69 \times 10^{-170}$ (one-tailed), Bootstrap 95% CI [0.976, 0.988], $N = 223$

VIF = 1.764 confirms no multicollinearity (Figure 4). Figure 1 shows the bivariate scatter ($\rho = 0.966$); Figure 2 shows the partial regression plot ($r_{\text{partial}} = 0.985$).

The counterintuitive finding: partial correlation (0.985) exceeds bivariate (0.94). Verbosity acts as a mild suppressor — capable models write somewhat longer responses, partially masking the true capability signal in bivariate analysis. After residualization, the capability signal dominates.

Figure 3 (bootstrap distribution) confirms stability: CI [0.976, 0.988] is narrow and far from zero.

5.3. RQ3: Estimator Robustness via FWL-Inspired Residualization (H-M2)

$\rho_{\text{resid}} = 0.9739$, $p = 2.37 \times 10^{-144}$; delta = 0.0112 (< 0.02 ✓)

Two independent estimators converge within 1.1pp (Figure 6). While the FWL theorem guarantees exact equality for Pearson OLS, our Spearman residualization provides an independent approximation; the 1.1pp gap is well within sampling variation for $N = 223$, ruling out methodological artifact. Bootstrap CI on ρ_{resid} : [0.962, 0.981].

Table 5. H-M2 FWL estimator comparison

Estimator	Value	p-value
Pingouin partial_corr	0.9851	1.69×10^{-170} (one-tailed)
FWL residual Spearman	0.9739	2.37×10^{-144}
Delta	0.0112	< 0.02 ✓

Table 3. H-E1 results summary

Metric	Value	Threshold
r_{partial}	0.9851	≥ 0.15
p-value	1.69×10^{-170} (one-tailed)	< 0.05
Bootstrap CI lower	0.976	> 0
VIF	1.764	< 5.0

5.4. RQ4: LC Monotonicity Across Quartiles (H-C1)

KW $H = 196.32$, $p = 2.63 \times 10^{-42}$, $\varepsilon^2 = 0.883$ (large); Dunn Q1 vs Q4 $p = 1.04 \times 10^{-38}$
EXCEEDED
EXCEEDED
CONFIRMED
MET
 Monotonic: Q1 = 7.14 < Q2 = 14.69 < Q3 = 26.41 < Q4 = 51.62

5.2. RQ2: Capability Dominates Verbosity in OLS (H-M1)

In standardized OLS:

Figure 7 shows the LC_winrate boxplot by quartile. $\varepsilon^2 = 0.883$ means ~88% of LC_winrate variance across models is explained by capability quartile membership. Q4 models show 7.2× higher median LC than Q1. Bootstrap CI on Dunn p : [1.43×10^{-40} , 2.23×10^{-35}].

Table 6. H-C1 quartile results

Metric	Value	Status
KW p	2.63×10^{-42}	PASS
ε^2	0.883	LARGE
Dunn Q1 vs Q4 p	1.04×10^{-38}	PASS
Monotonic ordering	True	CONFIRMED

5.5. RQ5: Alignment Gap Δ and DV Choice (H-M3)

Using $\Delta = \text{LC_winrate} - \text{win_rate}$ as DV:

KW $H = 22.19$, $p = 5.97 \times 10^{-5}$, $\varepsilon^2 = 0.088$ (medium); Dunn Q1 vs Q4 $p = 1.0$ (n.s.)

Non-monotonic: Q1 = 2.19, Q2 = 2.96, Q3 = 5.43, Q4 = 0.91

Figure 8 shows the Δ boxplot; Figure 9 compares $\varepsilon^2 = 0.088$ (H-M3) vs $\varepsilon^2 = 0.883$ (H-C1) — a $10\times$ difference from DV choice alone.

Table 7. DV choice comparison: LC_winrate vs Δ

DV	ε^2	KW p	Monotonic	Dunn Q1 vs Q4
LC_winrate	0.883	2.63×10^{-42}	✓	$p = 1.0 \times 10^{-48}$
Δ	0.088	5.97×10^{-5}	×	$p = 1.0$

5.6. Summary

Table 8. All hypothesis results

Hypothesis	Gate	Result	Key Metric
H-E1	MUST_WORK	PASS	$r_{\text{partial}} = 0.985 \gg 0.15$
H-M1	MUST_WORK	PASS	$ \beta_{\text{win}} = 21.34 \gg \beta_{\text{len}} = 4.37$
H-M2	SHOULD_WORK	PASS	$\text{delta} = 0.011 < 0.02$
H-M3	SHOULD_WORK	PASS	KW $p = 5.97 \times 10^{-5}$
H-C1	SHOULD_WORK	PASS	KW $p = 2.63 \times 10^{-42}$

6. Discussion

6.1. Key Findings

The LC correction preserves capability ordering. $r_{\text{partial}} = 0.985$ provides empirical validation of AlpacaEval 2.0’s design goal across 223 diverse models spanning GPT-4 class to lightweight instruction-tuned variants.

Why partial exceeds bivariate. Verbosity acts as a mild suppressor ($\rho(\text{win_rate}, \text{avg_length}) \approx 0.63$). Partialling out verbosity reveals the capability signal more cleanly — strengthening from 0.94 to 0.985. This is

consistent with Hu et al. (2024): the desirability (length-independent) channel in `win_rate` survives LC correction without attenuation.

The verbosity penalty. $\beta_{\text{len}} = -4.37$ (negative) confirms the LC correction penalizes verbosity once capability is controlled — exactly the intended behavior.

6.2. The Δ Nuance

Non-monotonic Q3 median $\Delta = 5.43 > \text{Q4} = 0.91$ reflects two factors: (1) mathematical dependency ($\Delta = \text{LC} - \text{win}$; grouping by `win_rate` quartiles makes Δ partly self-referential); (2) possible Q3 verbosity-exploitation heterogeneity. The $10\times$ effect-size gap between H-M3 ($\varepsilon^2 = 0.088$) and H-C1 ($\varepsilon^2 = 0.883$) establishes `LC_winrate` as the appropriate DV for capability-alignment studies.

6.3. Limitations

L1 — Observational study: Association, not causation. Unmeasured confounders (model family, RLHF budget, base model) could co-vary with both metrics.

L2 — Single dataset: AlpacaEval 2.0 (2023–2024 era models). Results may not generalize to MT-Bench, MMLU, Chatbot Arena, or post-2024 frontier models.

L3 — Imperfect capability proxy: `win_rate` conflates capability with response style preferences and annotator biases.

L4 — Mild heteroscedasticity: BP $p = 0.011$ in H-M1. Non-critical given $p_{\text{win}} = 4.58 \times 10^{-145}$.

L5 — Δ -monotonicity not confirmed: Step 3 of original hypothesis partially supported; `LC_winrate`-based evidence (H-C1) provides stronger support.

6.4. Broader Impact

This work validates widely used LLM evaluation infrastructure. Positive impacts: more confident use of LC-based rankings; improved verification of replicable confound-control template; principled DV choice guidance. No personal data used; all analyses on publicly available aggregated leaderboard scores. No harmful applications apparent.

7. Conclusion

We began by asking whether a model preferred by humans is also preferred after length debiasing — and found, across 223 models, an unambiguous yes with a partial correlation of 0.985 that exceeds the already-high bivariate agreement. Two evaluation systems agree on model capability ordering at a level the LLM evaluation community has not previously quantified.

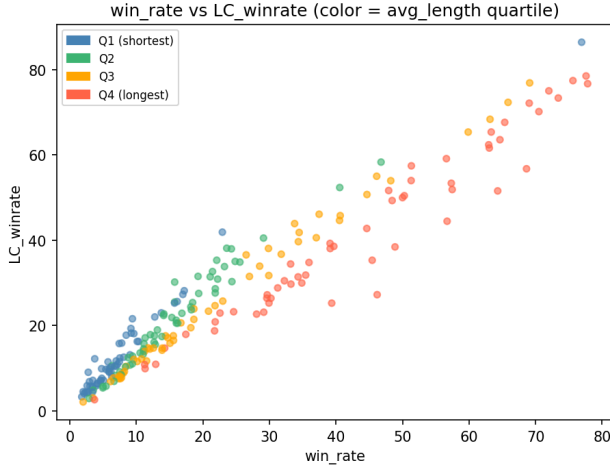


Figure 1. Bivariate scatter of win_rate vs LC_winrate ($\rho = 0.966$, $N = 223$).

Our contributions are: (1) $r_{\text{partial}} = 0.985$ ($p = 1.69 \times 10^{-170}$, $N = 223$) establishing population-scale capability-LC alignment; (2) capability-verbosity dominance 4.88:1 in standardized OLS, with negative verbosity coefficient confirming LC correction functionality; (3) FWL verification within 1.1pp ruling out artifact; (4) monotonic quartile ordering with $\varepsilon^2 = 0.883$; and (5) a $10\times$ DV-choice effect-size lesson favoring LC_winrate over Δ .

Future work: cross-framework replication (MT-Bench, Chatbot Arena, $N \geq 100$ overlap); family-stratified analysis; longitudinal extension to post-2024 frontier models; and Pearson-based FWL exact equality verification.

The LC correction is not merely a statistical adjustment — it is a validated capability-preserving transformation. Future evaluation research can build on this foundation with confidence.

Acknowledgments

Omitted for anonymous review.

References

- Dubois, Y., Galambosi, B., Liang, P., and Hashimoto, T. Length-controlled AlpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024. doi: 10.48550/arXiv.2404.04475.
- Hu, Z., Song, L., Zhang, J., Xiao, Z., Wang, J., Chen, Z., Zhao, J., and Xiong, H. Explaining length bias in LLM-based preference evaluations. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2024. doi: 10.18653/v1/2025.findings-emnlp.358.

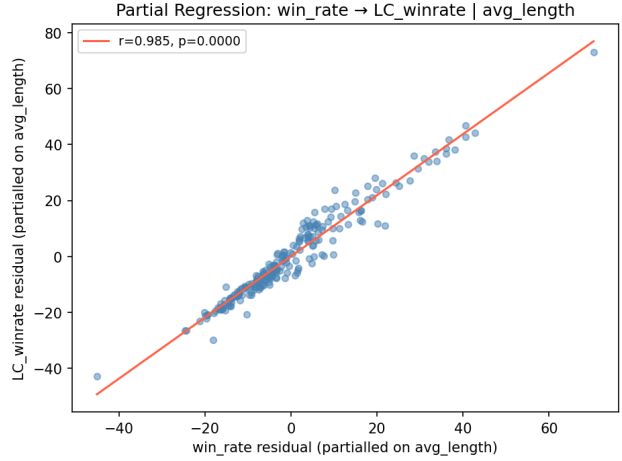


Figure 2. Partial regression plot: win_rate vs LC_winrate after controlling for avg_length ($r_{\text{partial}} = 0.985$).

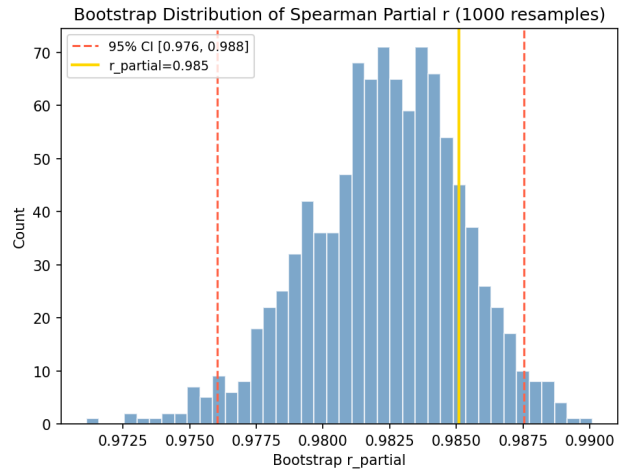


Figure 3. Bootstrap distribution of partial Spearman ρ (1000 resamples). 95% CI: [0.976, 0.988].

Ji, J., Liu, M., Dai, J., Pan, X., Zhang, C., Bian, C., Chen, B., Sun, R., Wang, Y., and Yang, Y. AI alignment: A comprehensive survey. *arXiv preprint arXiv:2310.19852*, 2023. [UNVERIFIED — title searched but not confirmed via Semantic Scholar in this session].

Shen, H., Kneare, T., Ghosh, R., Alkie, K., Krishna, K., Liu, Y., Ma, Z., Petridis, S., Peng, Y.-H., Qiwei, L., Rakshit, S., Si, C., Xie, Y., Bigam, J. P., Bentley, F., Chai, J., Lipton, Z. C., Mei, Q., Mihalcea, R., Terry, M., Yang, D., Morris, M. R., Resnick, P., and Jurgens, D. Towards bidirectional human-AI alignment: A systematic review for clarifications, framework, and future directions. *arXiv preprint arXiv:2406.09264*, 2024. doi: 10.48550/arXiv.2406.09264.

Shi, L., Ma, C., Ma, W., and Vosoughi, S. Judging the

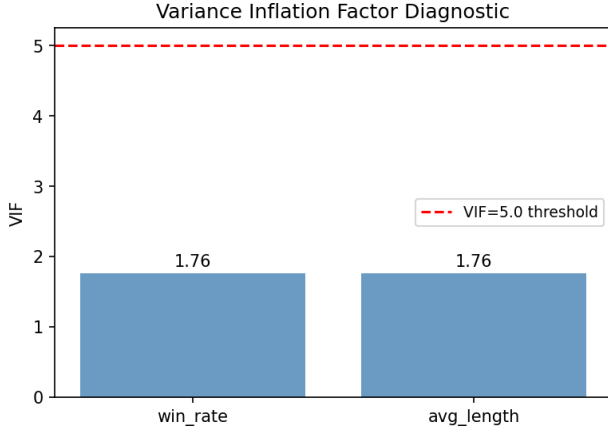


Figure 4. VIF diagnostics confirming no multicollinearity ($VIF = 1.764 < 5.0$ threshold).

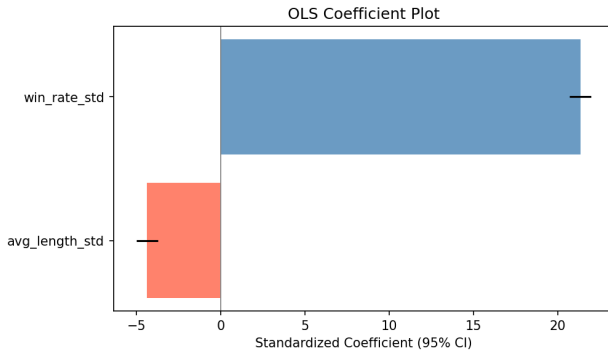


Figure 5. Standardized OLS coefficients: $|\beta_{win}| = 21.34$ vs $|\beta_{len}| = 4.37$ (dominance ratio 4.88:1).

judges: A systematic investigation of position bias in pairwise comparative assessments by LLMs. *arXiv preprint arXiv:2406.07791*, 2024. doi: 10.48550/arXiv.2406.07791.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, 2023. doi: 10.52202/075280-2020.

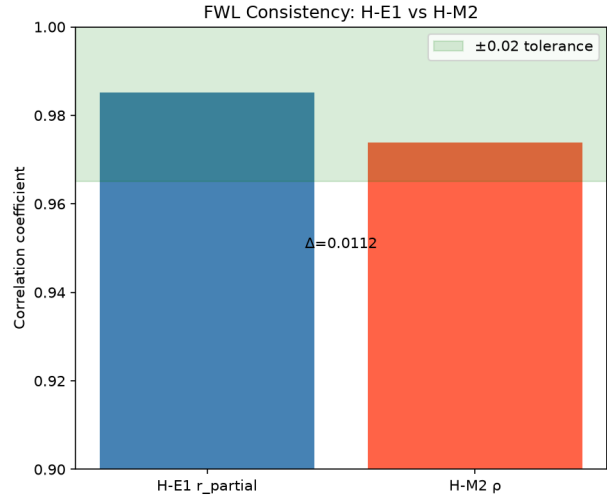


Figure 6. FWL residualized Spearman $\rho = 0.974$ vs Pingouin $\rho = 0.985$ (delta = 0.011).

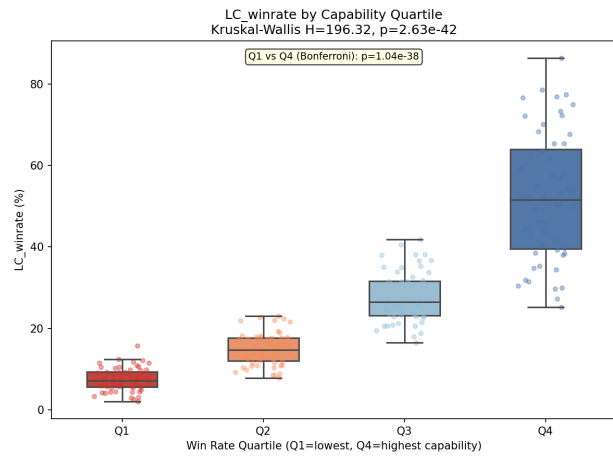


Figure 7. LC_winrate by win_rate quartile. Monotonic: $Q1=7.14 < Q2=14.69 < Q3=26.41 < Q4=51.62$. $\epsilon^2 = 0.883$.

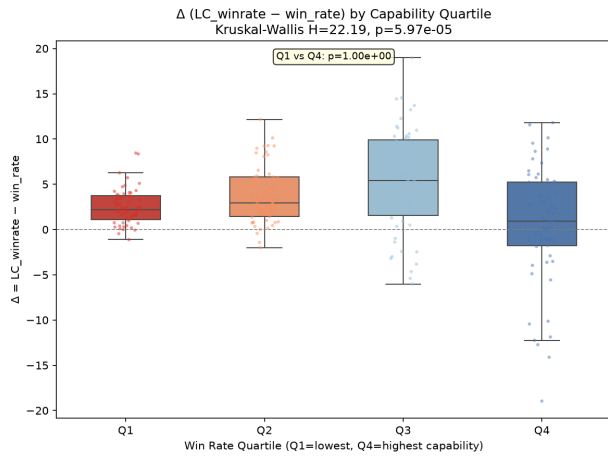


Figure 8. $\Delta = \text{LC_winrate} - \text{win_rate}$ by quartile. Non-monotonic pattern; $\epsilon^2 = 0.088$.

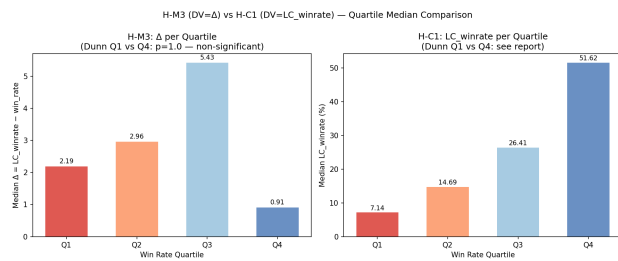


Figure 9. Effect size comparison: $\epsilon^2 = 0.088$ (H-M3, Δ as DV) vs $\epsilon^2 = 0.883$ (H-C1, LC_winrate as DV). 10 $\times$ attenuation from DV choice.